

SOLAR ROI & SAVINGS GUIDE

How to Calculate Your Solar Investment Returns | Gautam Solar 2024

Understanding Solar Economics

Solar power offers one of the best risk-adjusted returns available to Indian homeowners, businesses, and farmers. With electricity tariffs rising 5–8% annually and solar costs falling, the economics have never been more favourable. This guide helps you estimate your exact returns.

Key Financial Metrics

LCOE (Levelized Cost of Energy): Total lifetime cost of the solar system divided by total energy produced. For Gautam Solar TOPCon modules, typical LCOE is Rs 2.1–2.8/kWh — far below grid tariff.

Payback Period: Time to recover your initial investment from electricity savings. Typical range: 4–7 years for commercial, 5–8 years for residential.

ROI (Return on Investment): Annualized return over 25 years. Typical solar ROI in India: 15–25% depending on location and tariff.

NPV (Net Present Value): Present value of all future savings minus initial cost. Positive NPV confirms the investment is profitable.

Sample ROI Calculations by System Size

System Size	Segment	Approx. Cost (Rs)	Monthly Savings	Payback	25-yr Savings
3 kWp	Residential	1,80,000	Rs 1,800–2,400	6–7 years	Rs 4.5–6 Lakhs
5 kWp	Residential	2,80,000	Rs 3,000–4,000	6–7 years	Rs 7–9 Lakhs
10 kWp	Small Business	5,50,000	Rs 6,000–8,500	5–6 years	Rs 14–18 Lakhs
25 kWp	Commercial	13,00,000	Rs 15,000–22,000	5–6 years	Rs 35–48 Lakhs
100 kWp	Industrial	48,00,000	Rs 60,000–90,000	4–5 years	Rs 1.4–1.9 Cr
500 kWp	Utility/EPC	2,20,00,000	Rs 2.5–3.5 Lakhs	4–5 years	Rs 6–8 Cr

Solar Generation by Indian City (1kWp system)

City	State	Peak Sun Hours/Day	Units/Day (1kWp)	Units/Year (1kWp)
Jodhpur	Rajasthan	6.5 hrs	5.2–5.8 kWh	1900–2100
Ahmedabad	Gujarat	6.2 hrs	5.0–5.5 kWh	1820–2000
Delhi	Delhi NCR	5.5 hrs	4.4–4.9 kWh	1600–1790
Mumbai	Maharashtra	5.2 hrs	4.2–4.7 kWh	1530–1710
Chennai	Tamil Nadu	5.8 hrs	4.6–5.2 kWh	1680–1900
Bengaluru	Karnataka	5.4 hrs	4.3–4.8 kWh	1570–1750
Hyderabad	Telangana	5.6 hrs	4.5–5.0 kWh	1640–1820

Bhopal	Madhya Pradesh	5.7 hrs	4.6–5.1 kWh	1680–1860
Lucknow	Uttar Pradesh	5.3 hrs	4.2–4.7 kWh	1530–1710
Chandigarh	Punjab/Haryana	5.1 hrs	4.1–4.6 kWh	1500–1680

Gautam Solar | D-120-121, Okhla Industrial Area, Phase-1, New Delhi-110020
Email: info@gautamsolar.com | Phone: 1800-532-0800 | Web: www.gautamsolar.com